Angchen Xie

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Education

Carnegie Mellon University — Pittsburgh, PA
M.S. in Robotics (MSR) — *Aug. 2025 – May 2027 (expected)*
Advised by Guanya Shi and Max Simchowitz

- GPA: 4.17/4.00
- Courses: Introduction to Robot Learning; Optimal Control and Reinforcement Learning; Computer Vision; Advanced Control for Robotics; Mechanics of Manipulation

Shanghai Jiao Tong University — Shanghai, China
B.Eng. in Automation — *Sept. 2021 – June 2025*

- GPA: 4.01/4.30 | **Rank: 1/87**
- Undergraduate thesis: *Fault-Tolerant Control of Legged Robots under Multi-Sensor Fusion*

Research Interests

I work toward general-purpose robots that combine broad learned priors with grounded models of themselves and the physical world, enabling reliable and versatile behavior across tasks, embodiments, and environments.

Publications

* Equal contribution.

[1] **[Under review, CoRL 2026] Angchen Xie***, Nikhil Sobanbabu*, Ishayu Shikhare, Alan Wang, Max Simchowitz, Guanya Shi. *"FADA: Few-Shot Domain Adaptation via Dynamics Alignment for Humanoid Control".*
arXiv Website Thread

[2] **[RSS 2026]** Rishabh Madan, **Angchen Xie**, Samantha Saak, Andres Blanco, Dohyeok Lee, Sarah Grace Brown, Yunting Yan, Mark Zolotas, Jose Barreiros, Tapomayukh Bhattacharjee. *"TACTIC: Tactile and Vision Conditioned Contact-Centric Control for Whole-Arm Manipulation".*
arXiv Website

[3] **[RSS 2026]** Pablo Ortega-Kral*, Eliot Xing*, Arthur Bucker, Vernon Luk, Junseo Kim, Owen Kwon, **Angchen Xie**, Nikhil Sobanbabu, Yifu Yuan, Megan Lee, Deepam Ameria, Bhaswanth Ayapilla, Jaycie Bussell, Guanya Shi, Jonathan Francis, Jean Oh. *"RIO: Flexible Real-Time Robot I/O for Cross-Embodiment Robot Learning".*
arXiv Website Code

[4] **[CoRL 2025]** Rishabh Madan, Jiawei Lin, Mahika Goel, **Angchen Xie**, Xiaoyu Liang, Marcus Lee, Justin Guo, Pranav N. Thakkar, Rohan Banerjee, Jose Barreiros, Kate Tsui, Tom Silver, Tapomayukh Bhattacharjee. *"PrioriTouch: Adapting to User Contact Preferences for Whole-Arm Physical Human-Robot Interaction".*
arXiv Website

[5] **[IROS 2024] Angchen Xie**, Yeqiang Qian, Weihao Yan, Chunxiang Wang, Ming Yang. *"Non-repetitive: A Promising LiDAR Scanning Pattern".*
Paper Dataset

Research Experience

LeCAR Lab, Carnegie Mellon University — Pittsburgh, PA
Graduate Research Assistant — *Aug. 2025 – Present*
Advised by Guanya Shi and Max Simchowitz | humanoid control and cross-embodiment infrastructure

- Proposed and developed a Planner–IDM framework that freezes the planner and finetunes only the IDM on two minutes of target-domain rollouts, adapting to changes in terrain, payload, and actuator dynamics without rewards or demonstrations and raising real-robot success from 20% to 90%. *(FADA)*
- Co-developed an open-source cross-embodiment framework for control, teleoperation, and policy deployment, reducing observation-to-action latency to 130 ms versus 581 ms with LeRobot; led its humanoid integration, including latency benchmarking, policy training, and hardware deployment. *(RIO)*

RL^2 Lab, Shanghai Jiao Tong University — Shanghai, China
Undergraduate Research Assistant — *Jan. 2025 – June 2025*
Advised by Yue Gao | fault-tolerant locomotion for legged robots

- Developed a multi-sensor representation-learning controller that uses contrastive fault embeddings to adapt hexapod locomotion to joint failures, improving fault diagnosis from 78.6% to 91.2% and maintaining locomotion under up to three simultaneous joint failures after Isaac Gym-to-MuJoCo transfer. *(Undergraduate thesis)*

EmPRISE Lab, Cornell University — Ithaca, NY
Research Intern — *July 2024 – Dec. 2024*
Advised by Tapomayukh Bhattacharjee | contact-rich whole-arm physical human–robot interaction

- Developed the learned latent dynamics model, which fuses tactile force, RGB, and depth to drive a contact-centric whole-arm MPC controller, and built its simulation evaluation pipeline, reaching 79.5% success on simulated bed bathing versus 51.6% for DreamerV3 with 65% fewer force violations. *(TACTIC)*
- Built the Unity bed-bathing environment behind a learning-to-rank framework for conflicting force preferences, instrumenting the human model with force-sensor colliders and contributing to the ranking algorithm that significantly reduced force violations versus fixed and heuristic prioritization ($p < 0.05$). *(PrioriTouch)*

CyberC3 Intelligent Vehicle Lab, Shanghai Jiao Tong University — Shanghai, China
Undergraduate Research Assistant — *Nov. 2022 – June 2024*
Advised by Yeqiang Qian and Ming Yang | LiDAR perception for intelligent vehicles

- Built *Repetitive-or-not*, a controlled CARLA benchmark for repetitive versus non-repetitive LiDAR scanning, and showed that non-repetitive scanning captures up to 59.6% more objects, particularly smaller ones such as pedestrians, while quantifying the resulting domain gap in 3D object detection. *(Non-repetitive)*

Service

Reviewer, Conference on Robot Learning (CoRL)	2026
Reviewer, Robotics: Science and Systems (RSS)	2026
Reviewer, IEEE Robotics and Automation Letters (RA-L)	2025, 2026

Skills

Programming	Python, C/C++, MATLAB, LaTeX
Frameworks	PyTorch, ROS/ROS2, Isaac Gym, Isaac Lab, MuJoCo, Gazebo, CARLA, Unity
Robots	Unitree G1, Unitree Go1/Go2, AgiBot G1, Booster T1, Kinova

Honors and Awards

Outstanding Graduate Award, Shanghai Jiao Tong University	2025
China National Scholarship (top 0.2%)	2024
First Prize Academic Scholarship, Shanghai Jiao Tong University	2024
National First Prize, China Undergraduate Electronics Design Contest (top 1%)	2023
First Prize, China Undergraduate Mathematical Contest in Modeling (top 2%)	2022